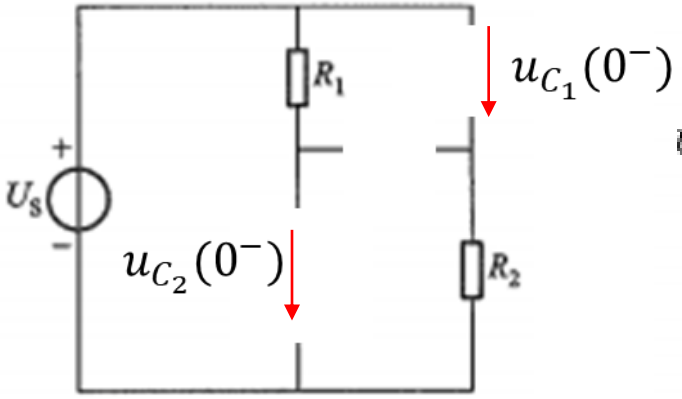
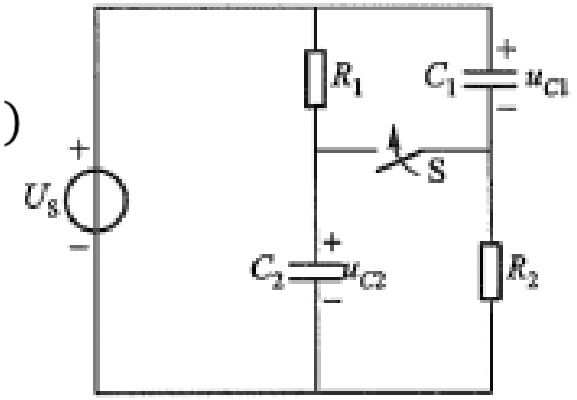
7.5　在题图 7.5 所示电路中 $C_1=1\ \mathrm{F}$，$C_2=2\ \mathrm{F}$，$R_1=1\ \Omega$，$R_2=2\ \Omega$，$U_S=9\ \mathrm{V}$，开关 S 打开已久，$t=0$ 时将 S 闭合，试求 $u_{C1}(0_+)$、$u_{C2}(0_+)$。





题图 7.5

$$u_{C_1}(0^-)=u_{C_2}(0^-)=9$$

$$u_{C_1}(0^+)+u_{C_2}(0^+)=9$$

$$-C_1u_{C_1}(0^-)+C_2u_{C_2}(0^-)=-C_1u_{C_1}(0^+)+C_2u_{C_2}(0^+)$$

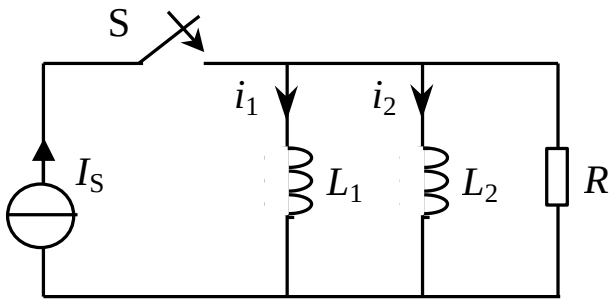
$$-1\times9+2\times9=-\left[9-u_{C_2}(0^+)\right]+2u_{C_2}(0^+)$$

$$-1\times9+2\times9=-\left[9-u_{C_2}(0^+)\right]+2u_{C_2}(0^+)\longrightarrow u_{C_2}(0^+)=6$$

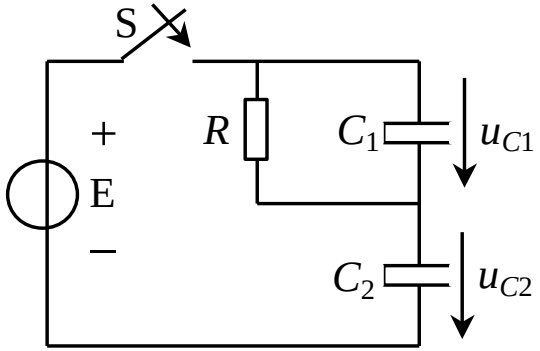
$$u_{C_1}(0^+)=3$$

$$u_{C_1}(\infty)=3\qquad u_{C_2}(\infty)=6\qquad \tau=\left(\frac{1\times2}{1+2}\right)(C_1+C_2)$$

图电路中，I_S=6A，L_1=1H，L_2=2H，R=1Ω，$i_1(0–)$=1A，$i_2(0–)$=2A，则S闭合后，$i_2(0+)$=__2__A，$i_2(\infty)$=__3__A。



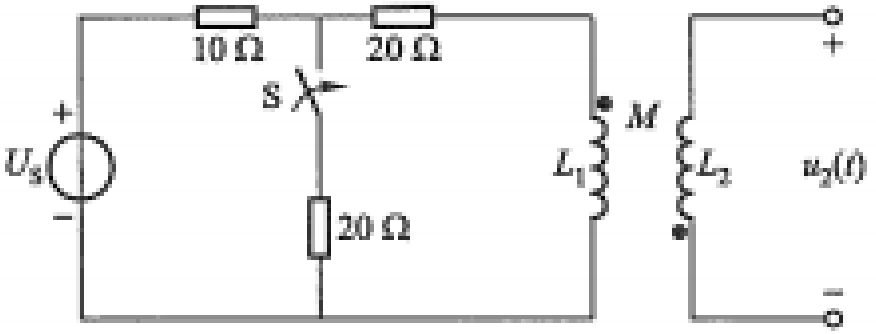
图电路中，E=10V，R=1Ω，C_1=1F，C_2=2F，$u_{C1}(0–)$=3V，$u_{C2}(0–)$=4V，则S闭合后，$u_{C2}(0+)$=__5__V，$u_{C2}(\infty)$=__10__V，$u_{C1}(\infty)$=__0__V。



$$u_{C1}(0^+)+u_{C2}(0^+)=10$$

$$-C_1u_{C1}(0^+)+C_2u_{C2}(0^+)=-1\times3+2\times4$$

7.6　在题图 7.6 所示电路中，S 闭合前电路已达稳态，$t=0$ 时 S 闭合，求：$t>0$ 时的 $u_2(t)$，其中 $U_S=30\ \text{V}$，$L_1=800\ \text{mH}$，$L_2=200\ \text{mH}$，$M=50\ \text{mH}$。



题图 7.6

$$i_{L_1}(0^-)=\frac{U_S}{10+20}=1A=i_{L_1}(0^+)$$

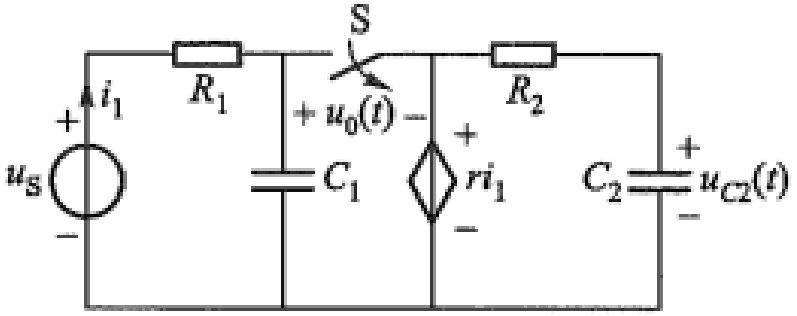
$$i_{L_1}(\infty)=\frac{U_S}{10+20/2}0.5=0.75A$$

$$\tau=\frac{L}{20+\dfrac{10\times 20}{10+20}}=\frac{3}{100}(s)$$

$$i_{L_1}(t)=0.75+0.25e^{-\frac{100t}{3}}A$$

$$u_2(t)=-M\frac{di_{L_1}(t)}{dt}=-0.05\times 0.25\left(-\frac{100}{3}\right)e^{-\frac{100t}{3}}=\frac{1.25}{3}e^{-\frac{100t}{3}}V$$

7.12 题图 7.12 所示电路中，已知 $R_1=20\ \Omega$，$R_2=10\ \Omega$，$r=20\ \Omega$，$C_1=C_2=0.1\ \mathrm{F}$，$u_S=12\ \mathrm{V}$，开关 S 闭合已久，试求 S 打开后其两端的电压 $u_0(t)$ 与 $u_{C2}(t)$。



题图 7.12

$$u_{C_1}(\infty)=12 \qquad u_{C_1}(0^+)=6 \qquad \tau=20\times0.1$$

$$u_{C_1}(t)=\left(12-6e^{-0.5t}\right)V$$

$$i_1(t)=0.1\times6\times0.5e^{-0.5t}$$

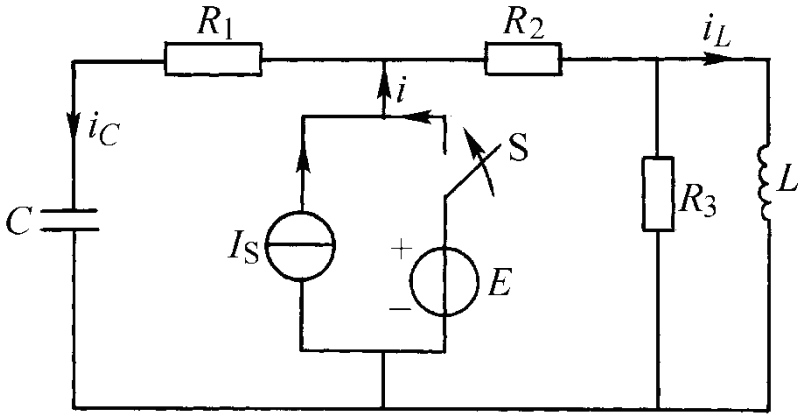
$$RC\frac{du_{C_2}}{dt}+u_{C_2}=20\times0.3e^{-0.5t} \qquad u_{C_2}(0^+)=6 \qquad \tau=10\times0.1$$

$$R_2C_2A(-0.5)e^{-0.5t}+Ae^{-0.5t}=20\times0.3e^{-0.5t} \qquad A=12$$

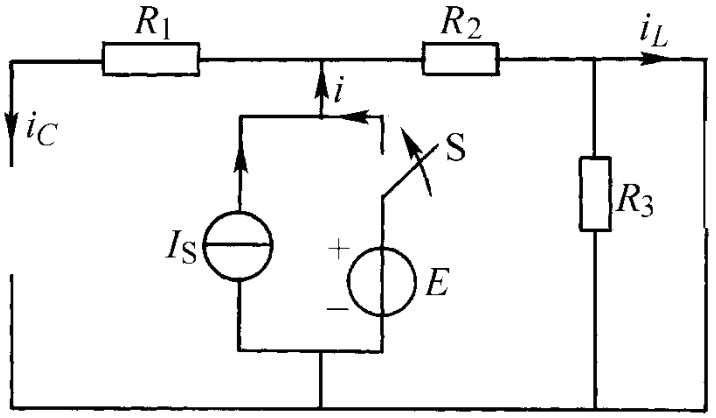
$$u_{C_2}(t)=\left(12e^{-0.5t}+(6-12)e^{-t}\right)V$$

7-12 $\quad u_{C_1}(t)=\left(12-6e^{-0.5t}\right)V \qquad u_{C_2}(t)=\left(12e^{-0.5t}-6e^{-t}\right)V$

7-14 题图 6-30 所示电路中，已知$R_1=100\Omega, R_2=30\Omega, R_3=60\Omega$，$C=100\mu F$，$L=100mH$，$I_S=0.5A$ $E=30V$，开关 S 原为断开，电路已达稳态，现于 $t=0$ 时闭合 S，求 $i_C(t)$ 及 $i(t)$。



解：



$$i_L(t)=1+(0.5-1)e^{-200t}$$

$$u_C(t)=30+(15-30)e^{-100t}$$

$$i_C(t)=\frac{30-\left(30+(15-30)e^{-100t}\right)}{100}$$

$$i=i_C+\frac{L}{R_3}\frac{di_L}{dt}+i_L=1+0.15e^{-100t}-\frac{1}{3}e^{-200t}$$

$$i_L(0^-)=I_s=0.5A$$

$$u_C(0^-)=R_2I_s=30\times0.5=15V$$

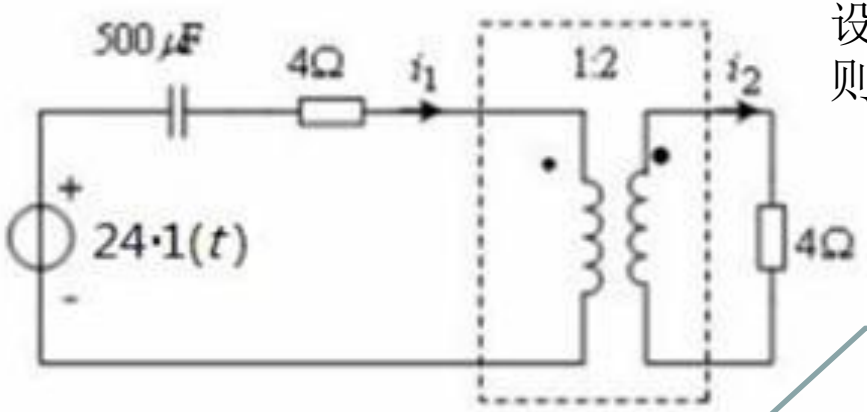
$$u_C(\infty)=E=30V$$

$$i_L(\infty)=\frac{E}{R_2}=1A$$

$$\tau_C=R_1C=10^{-2}s$$

$$\tau_L=\frac{L}{R_2/\!/R_3}=5\times10^{-3}s$$

图示含理想变压器电路，已知电源电压为$u_s(t)=24\cdot 1(t)V$，求$i_1(t)$和$i_2(t)$



设理想变压器电压电流为关联参考方向，则其特性为：

$$u_1 = nu_2$$

$$i_1 = \frac{1}{n} i_2$$

A选项应该将i2加个负号

A. $i_1 = 4.8e^{-400t}1(t)$； $i_2 = -2.4e^{-400t}1(t)$

B. $i_1 = 2e^{-250t}1(t)$； $i_2 = 4e^{-250t}1(t)$

C. $i_1 = -4.8e^{-400t}1(t)$； $i_2 = 2.4e^{-400t}1(t)$

D. $i_1 = 1.2e^{-100t}1(t)$； $i_2 = -2.4e^{-100t}1(t)$

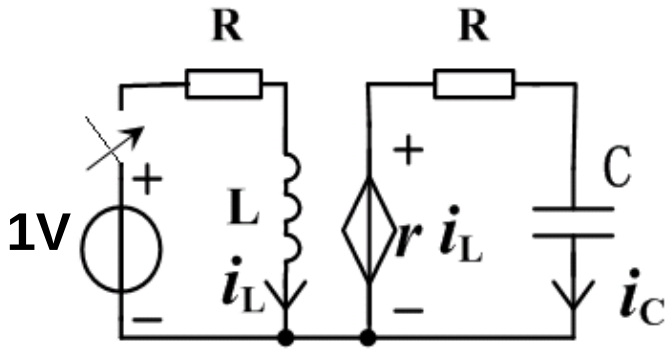
解：将负载4欧姆等效到理想变压器1次侧为1欧姆

$$\tau = (4+1)500\times10^{-6} \quad u_C(0^-) = 0, \quad u_C(\infty) = 24 \qquad u_C(t) = 24\left(1-e^{-400t}\right)$$

$$i_1 = C\frac{du_C}{dt} = 500\times10^{-6}\times 24\left(400e^{-400t}\right) = 4.8e^{-400t}$$

$$i_2 = \frac{1}{2} i_1 = 2.4e^{-400t}$$

例题9: 如图电路，$u_C(0^-)=0, i_L(0^-)=0$



试求 $i_C(t)$

解：

零状态响应

左侧**RL**电路：

$$i_L(t)=i_L(\infty)+(i_L(0^+)-i_L(\infty))e^{-\frac{R}{L}t}=\frac{1}{R}(1-e^{-\frac{R}{L}t})$$

右侧**RC**电路：

$$RC\frac{du_C}{dt}+u_C=\gamma\frac{1}{R}(1-e^{-\frac{R}{L}t})$$

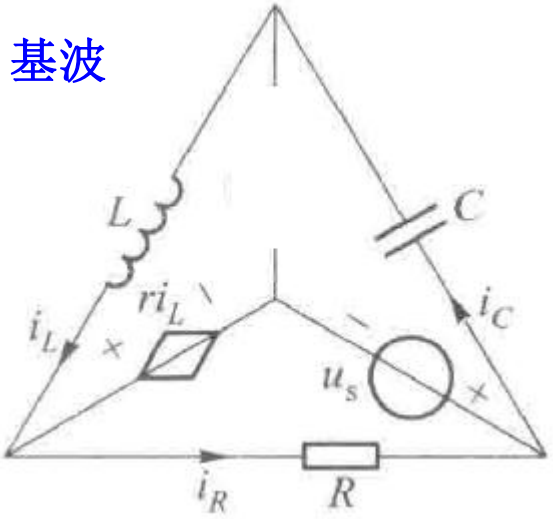
强制分量

设特解为 $u_{Cp}=Ae^{-\frac{R}{L}t}+B$

$$u_C(t)=u_{Cp}+(u_C(0^+)-u_{Cp}(0^+))e^{-\frac{1}{RC}t}$$

6.8 题图 6.8 所示电路，已知：$u_s(t)=30\sqrt{2}\sin\omega t$ V，$i_s(t)=3\sqrt{2}\sin(2\omega t+45°)$ A，且 $\gamma=30\ \Omega$，$R=10\ \Omega$，$\omega L=1/(\omega C)=20\ \Omega$。求 i_L、i_C、i_R 和电压源电流源的平均功率。

基波

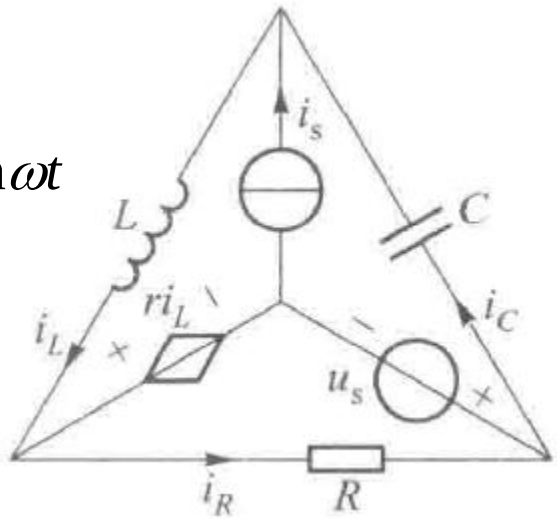


$$i_R=0$$

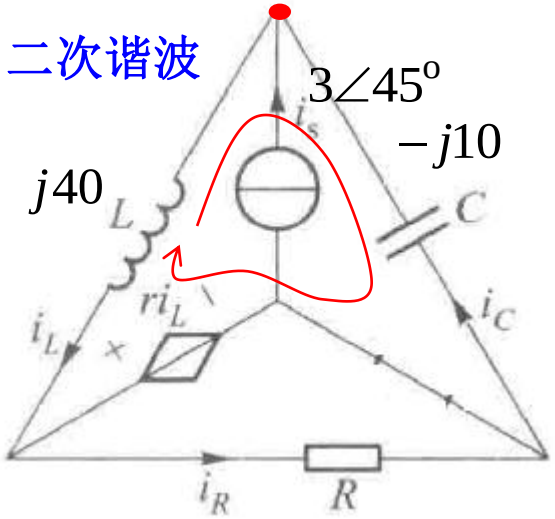
$$ri_L=u_s \qquad i_L=\frac{u_s}{r}=i_C=\sqrt{2}\sin\omega t$$

$$P_{u_s}=\frac{1}{T}\int_0^T u_s i_L\,dt=30W \quad \text{发出}$$

$$P_{i_s}=0$$



二次谐波



$$3\angle 45^o+\dot{I}_C=\dot{I}_L$$

$$30\dot{I}_L+\dot{I}_C(-j10)+\dot{I}_L j40=0$$

$$j30\angle 45^o-j10\dot{I}_L+30\dot{I}_L+\dot{I}_L j40=0\Rightarrow \dot{I}_L=\frac{-j30\angle 45^o}{30+j30}=\frac{-j}{\sqrt{2}}$$

$$\dot{I}_C=\dot{I}_L-3\angle 45^o=-\frac{1}{\sqrt{2}}(3+j4) \qquad \dot{I}_R=\frac{r\dot{I}_L}{R}=3\dot{I}_L=\frac{-j3}{\sqrt{2}}$$

吸收

$$P_{i_s}=Re\left\{-j10\dot{I}_C\times 3\angle -45^o\right\}=Re\left\{j\frac{30}{\sqrt{2}}(3+j4)\frac{\sqrt{2}}{2}(1-j)\right\}=-15W$$

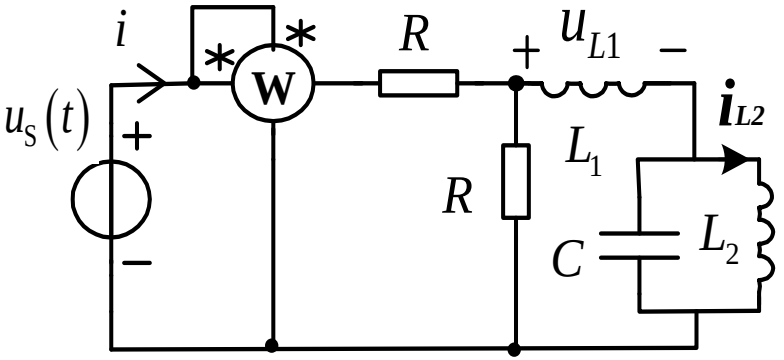
实际为发出**15W**

电容电压

已知 $R=1\Omega$，$C=5\,\mathrm{mF}$，$L_1=\frac{5}{3}\,\mathrm{mH}$，$L_2=5\mathrm{mH}$，

$u_S(t)=10+20\sqrt{2}\sin(200t+30^\circ)+10\sqrt{2}\sin(400t-60^\circ)\,\mathrm{V}$，求电流 $i(t)$，$u_{L1}(t)$，

$i_{L2}(t)$及功率表读数。



直流：$i^{(0)}=10A$　$i_{L_2}^{(0)}=1A$

$P^{(0)}=100W$　$u_{L_1}^{(0)}=0V$

基波：L_2-C并联谐振　$\dot{I}^{(1)}=10\angle 30^\circ A$　$\dot{I}_{L_2}^{(1)}=10\angle -60^\circ A$　$\dot{U}_{L_1}^{(1)}=0V$

$P^{(1)}=200W$

二次：L_2-C-L_1串联谐振

$\dot{I}^{(2)}=10\angle -60^\circ A$　$\dot{I}_{L2}^{(2)}=\frac{10}{3}\angle 120^\circ A$　$\dot{U}_{L_1}^{(2)}=j\frac{2}{3}\times 10\angle -60^\circ A$

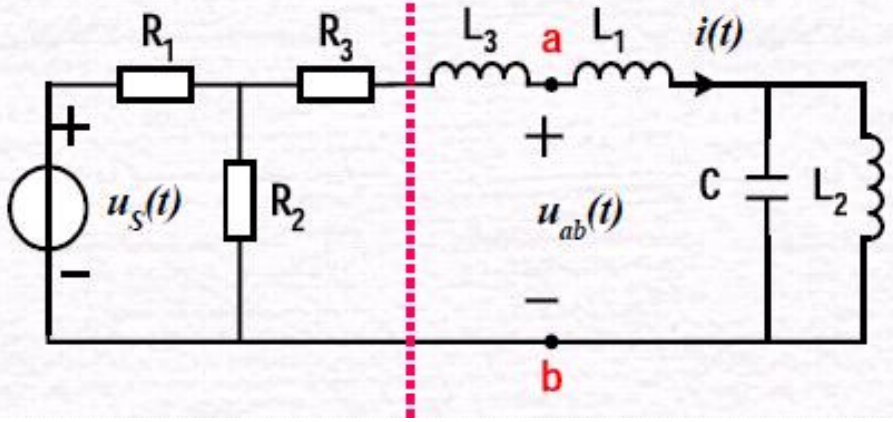
$P^{(2)}=100W$

$$i(t)=10+10\sqrt{2}\cos(200t+30^\circ)+10\sqrt{2}\cos(400t-60^\circ)\,A$$

例2：已知：$R_1=R_2=20\,\Omega$，$R_3=30\,\Omega$，$L_2=1\text{mH}$，$L_3=1.5\ \text{mH}$，

$$u_S(t)=320+200\sqrt{2}\sin 10000t+300\sqrt{2}\sin 20000t\ (V)$$

要使 $u_{ab}(t)=100\sqrt{2}\sin 10000t\ V$ 求：L_1，C，$i(t)$



解：戴维南等效电路

$$u_{Sd}(t)=u_{Sd}(0)+u_{Sd}(1)+u_{Sd}(2)$$

$$u_{Sd}(0)=\frac{u_S(0)}{R_1+R_2}R_2=160\ (V)$$

$$\dot{U}_{Sd}(1)=\frac{\dot{U}_S(1)}{R_1+R_2}R_2=\frac{1}{2}\dot{U}_S(1)=100\angle 0^\circ\ (V)$$

$$\dot{U}_{Sd}(2)=\frac{\dot{U}_S(2)}{R_1+R_2}R_2=\frac{1}{2}\dot{U}_S(2)=150\angle 0^\circ\ (V)$$

$$u_{Sd}(t)=160+100\sqrt{2}\sin 10000t+150\sqrt{2}\sin 20000t\ (V)$$

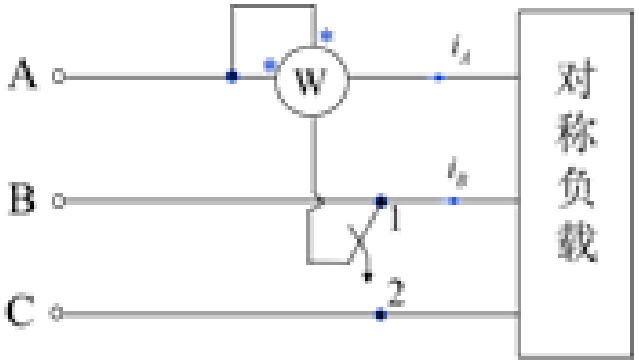
1/3mH, 10uF

6.19 对称三相电路，如题图6.19所示，已知相电压为

$$u_{\mathrm{A}} = 220\sqrt{2}\sin\omega t + 30\sqrt{2}\sin 3\omega t\ \mathrm{V}$$

负载Y接，每相由R、L组成，R=20Ω，ωL=40Ω

求线电流、开关在位置1和2的瓦特计读数，并求负载消耗的总功率。



$$\left|\frac{220}{R+j\omega L}\right| = 4.91A$$

$$\dot{I}_A = \frac{220}{R+j\omega L} = 4.91\angle -63.43^\circ A$$

$$\dot{U}_{AB} = 220\sqrt{3}\angle 30^\circ V \qquad \dot{U}_{AC} = 220\sqrt{3}\angle 30^\circ \angle 120^\circ \angle -180^\circ V$$

$$W_1 = 220\times 4.91\cos\left(30^\circ + 63.43^\circ\right)W \qquad W_2 = 220\times 4.91\cos\left(-30^\circ + 63.43^\circ\right)W$$

6.20 对称三相四线制，Y接电势含有基波和三次谐波，Y接负载为纯电阻，已知中线电阻为R_N=4Ω，中线电流9A，线电压为380V，三相电路总功率为6966W，求负载电阻。

$$3\frac{U_3}{R_L+3\times 4}=9(A)$$ 0序每相电流

0序三相总功率　　基波总功率

$$3\left(\frac{U_3}{R_L+3\times 4}\right)^2 R_L+\sqrt{3}380\frac{220}{R_L}+9^2\times 4=6966(W)$$

中线电阻功率

$$27R_L+\sqrt{3}380\frac{220}{R_L}=6642$$

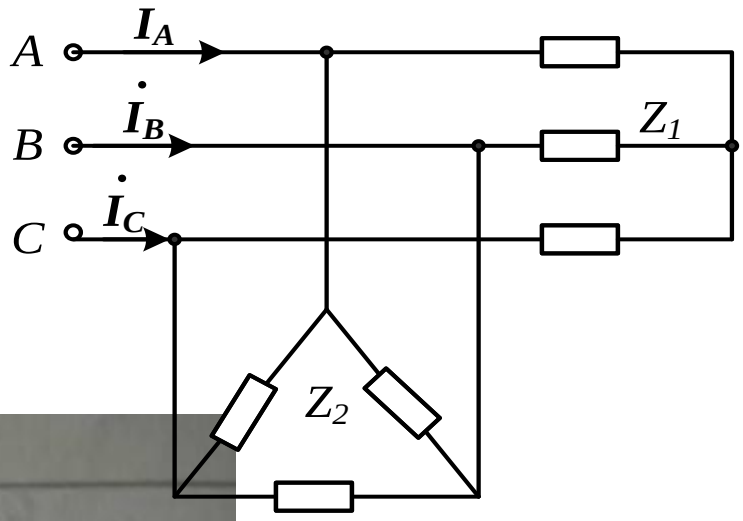
$$R_L=\frac{6642\pm\sqrt{6642^2-4\times 27\times\sqrt{3}\times 380\times 220}}{54}=\begin{cases}221.82\\24.17\end{cases}(\Omega)$$

5.50 线电压100V，星形负载消耗功率$P_1 = 60W$，功率因数$\cos\varphi_1 = 0.5$（$\varphi_1 > 0$），三角形负载消耗功率$P_2 = 45W$，$\cos\varphi_2 = 0.6$（$\varphi_2 < 0$），试求电源电流相量

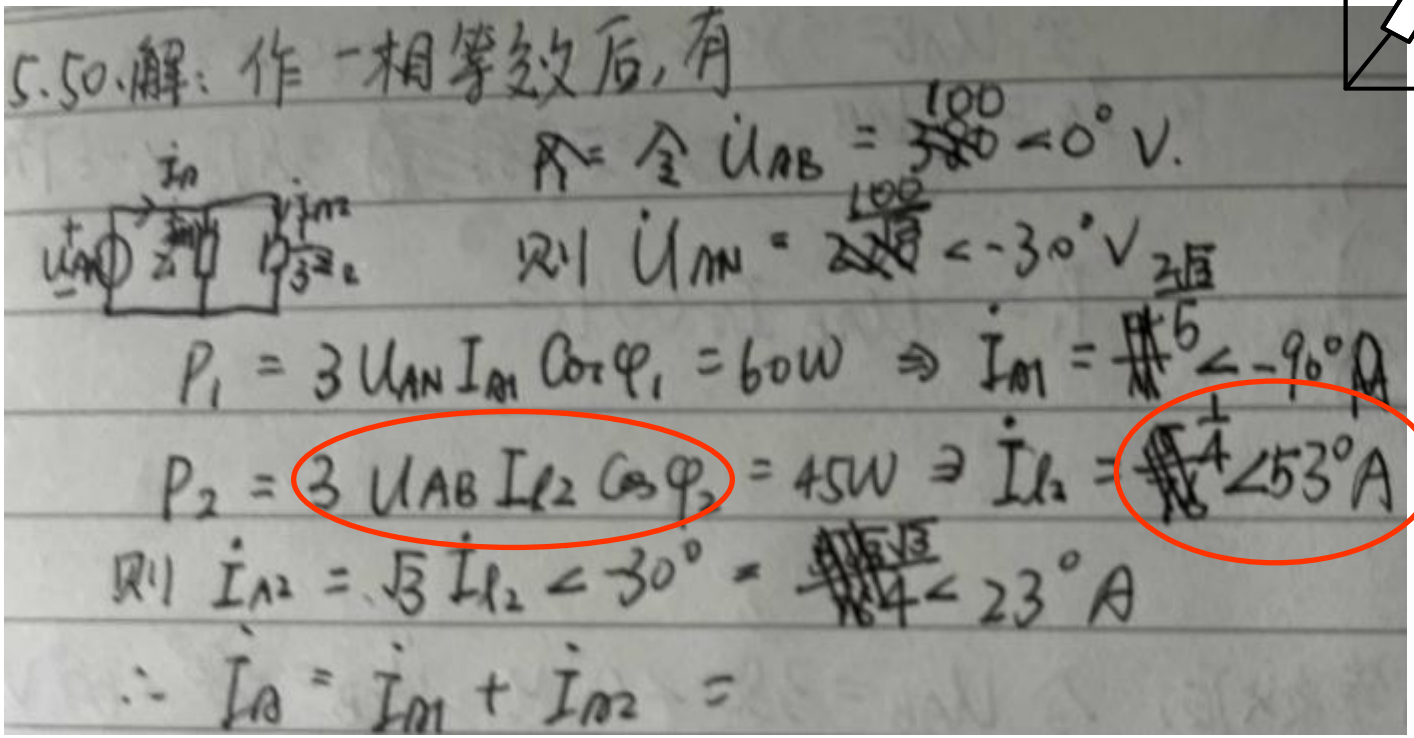
$$\dot{I}_A = \frac{2\sqrt{3}}{5}\angle -90^\circ + \frac{\sqrt{3}}{4}\angle 23^\circ$$

$$= 0.3985 - j0.5236$$

$$= 0.658\angle -52.7^\circ$$



功率因数角
=阻抗角



方法一

$$P = P_1 + P_2$$

$$Q = P_1 \frac{\sqrt{1-0.5^2}}{0.5} - P_2 \frac{\sqrt{1-0.6^2}}{0.6}$$

$$\tilde{S} \neq \sqrt{3}\dot{U}_\ell \dot{I}_\ell^* \quad \left|\tilde{S}\right| = \sqrt{3}U_\ell I_\ell$$

$$I_\ell = \frac{\sqrt{(60+45)^2 + \left(60\frac{\sqrt{0.75}}{0.5} - 45\times\frac{4}{3}\right)^2}}{\sqrt{3}100} = \frac{\sqrt{11025+1928.9}}{\sqrt{3}100} = 0.657$$

$$tg\varphi = \frac{Q}{P} = \frac{60\frac{\sqrt{0.75}}{0.5} - 45\times\frac{4}{3}}{60+45} = \frac{43.92}{105} \Rightarrow \varphi = 22.69^\circ$$

方法二

设A相电压为参考

$$\dot{I}_{AB}=\frac{\sqrt{3}U_A\angle\left(0^\circ+30^\circ\right)}{Z_\Delta}$$

三角形负载等效为Y后，其线电流与相电压的夹角是Δ负载的阻抗角或功率因数角

$$\dot{I}_{A\Delta}=\dot{I}_{AB}\sqrt{3}\angle-30^\circ=\frac{\sqrt{3}U_A\angle\left(0^\circ+30^\circ\right)\sqrt{3}\angle-30^\circ}{Z_\Delta}=\frac{\sqrt{3}U_A\angle0^\circ\sqrt{3}}{Z_\Delta}=I_\ell\angle-\varphi_\Delta$$

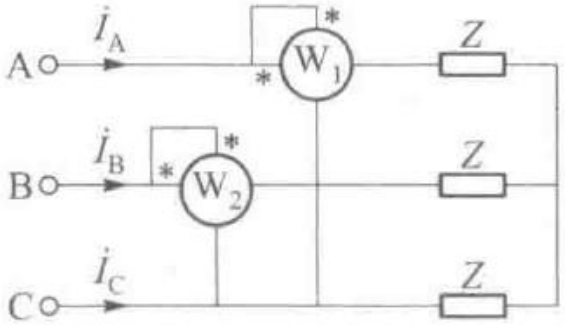
$$\dot{I}_A=\frac{P_1}{\sqrt{3}100\times0.5}\angle-60^\circ+\frac{P_2}{\sqrt{3}100\times0.6}\angle\cos^{-1}0.6$$

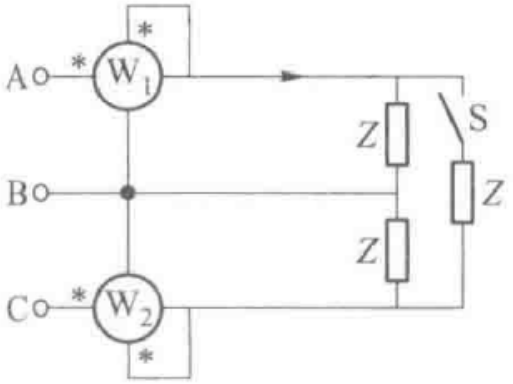
$$=0.346\left(1-j\sqrt{3}\right)+0.433\left(0.6+j0.8\right)$$

$$=0.6058-j0.2528=0.656\angle-22.65^\circ$$

5.56 题图 5.56 所示对称三相负载，已知线电压为 380 V，$Z=(8+j6)\ \Omega$，求各功率表的读数。

5.57 在题图 5.57 所示对称三相电路中，线电压 380 V，$Z=(8+j6)\ \Omega$，(1) 求线电流，两功率表读数，并利用此两读数求总有功功率和无功功率。(2) 若开关 S 断开，求两功率表读数。





$$W_1=U_lI_l\cos(30-\varphi)$$

$$W_2=U_lI_l\cos(30+\varphi)$$

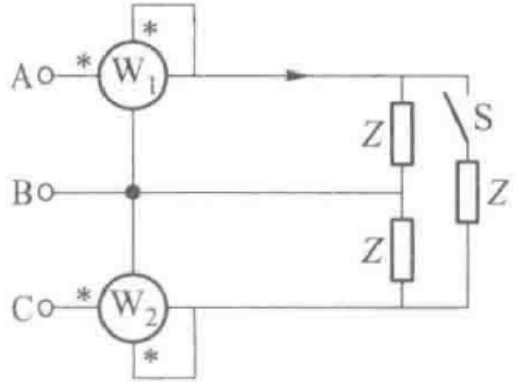
$$U_l=380\quad I_l=\frac{220}{\sqrt{8^2+6^2}}\quad \varphi=tg^{-1}\frac{6}{8}$$

(1)

$$W_1=U_lI_l\cos(30+\varphi)\qquad P=W_1+W_2$$

$$W_2=U_lI_l\cos(30-\varphi)\qquad Q=W_2-W_1$$

$$U_l=380\quad I_l=\frac{220}{\sqrt{\left(\frac{8}{3}\right)^2+2^2}}\quad \varphi=tg^{-1}\frac{6}{8}$$



（2）

$$W_1 = U_{AB}\left|\frac{\dot{U}_{AB}}{Z}\right|\cos\left(0^o+\varphi\right)=\left(\frac{380}{|Z|}\right)^2\times 8$$

$$W_2 = \left|-U_{AB}\angle -120^o\right|\left|\frac{-U_{AB}\angle -120^o}{Z}\right|\cos\left(\varphi\right)$$

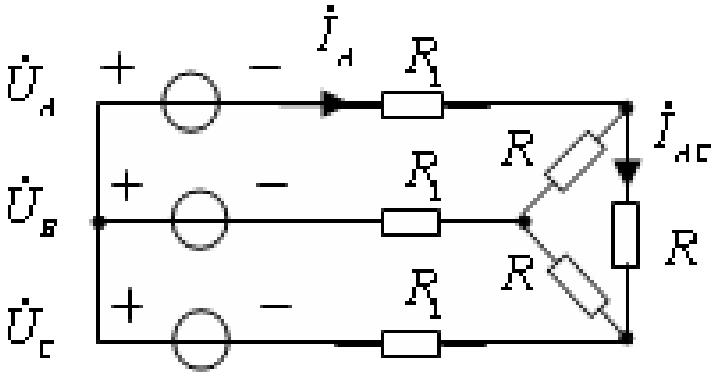
$$\varphi = tg^{-1}\frac{6}{8}$$

例： 图示对称三相电路中已知 $\dot{U}_A = 220\angle 0°\,\mathrm{V}$

已知 R = 300Ω， R_1 = 10Ω，则线电流

$\dot{I}_A$ =_____A ，负载相电流

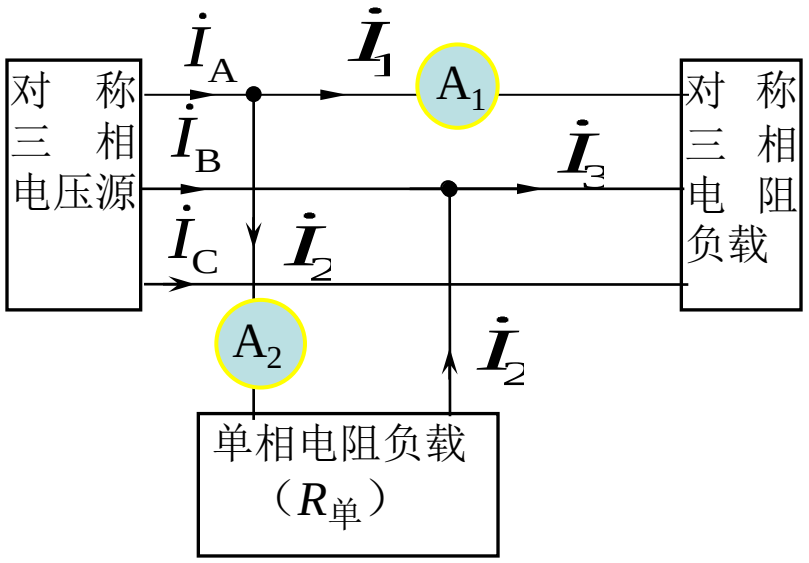
$\dot{I}_{AC}$ = _____A.



$$\dot{I}_A = \frac{\dot{U}_A}{R_1 + \dfrac{R}{3}} = 2\angle 0^\circ \times(-1)$$

$$\dot{I}_{AC} = \left(\frac{\dot{U}_A}{R_1 + \dfrac{R}{3}} / \sqrt{3} \right) \angle 30^\circ \angle 120^\circ \angle -180^\circ = \frac{2}{\sqrt{3}} \angle -30^\circ \times(-1)$$

[补充7.7]图示电路电流表的读数均为2A，求电流 I_A、I_B和 I_C。



解：设 $\dot{I}_1 = 2\angle 0°\,\mathrm{A}$

则 $\dot{I}_2 = \dfrac{\dot{U}_{AB}}{R_{单}} = 2\angle 30°\,\mathrm{A}$

$$\dot{I}_A = \dot{I}_1 + \dot{I}_2 = (2 + \sqrt{3} + \mathrm{j})\,\mathrm{A}$$

$$I_A = 3.61\,\mathrm{A}$$

$$\dot{I}_B = \dot{I}_3 - \dot{I}_2 = [(2\angle -120°) - 2\angle 30°]\,\mathrm{A}$$

$$I_B = 3.86\,\mathrm{A}$$

$$I_C = I_1 = 2\,\mathrm{A}$$

图示电路为380 / 220 伏三相四线制供电系统, 接有两组对称三相负载, 已知：$Z = 40 + j30\Omega$ (其余参数如图所示).

试求：

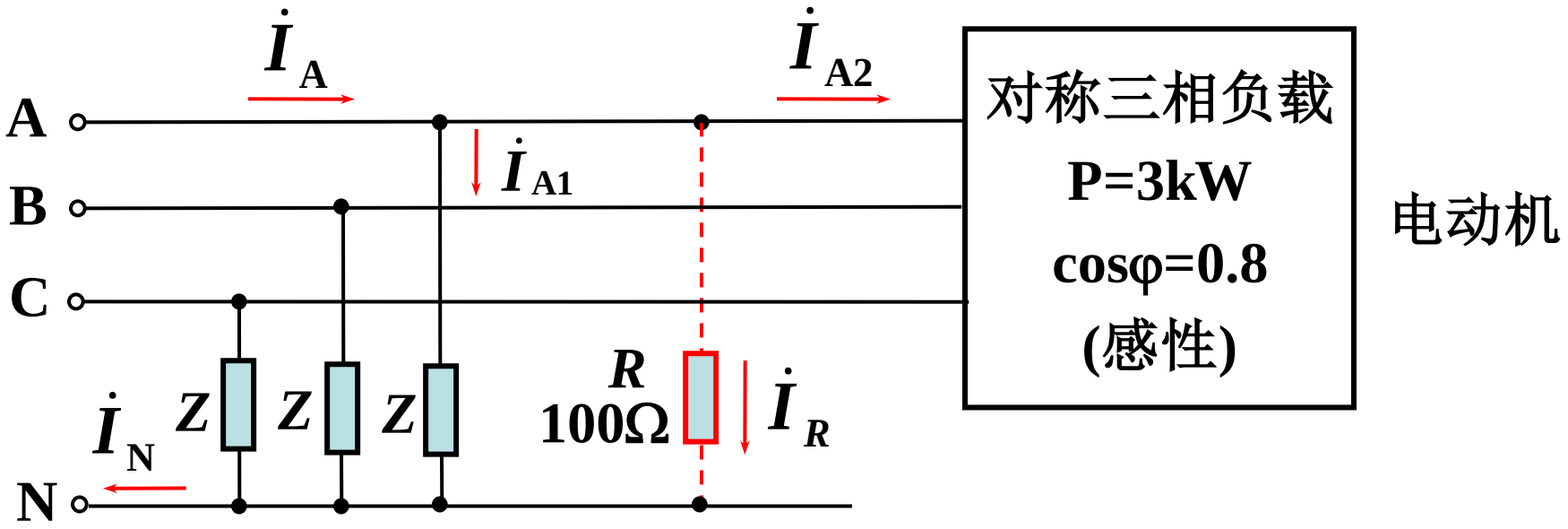
(1) 三个线电流 $\dot{I}_A$, $\dot{I}_{A1}$, $\dot{I}_{A2}$和中线电流$\dot{I}_N$;

(2) 计算三相电源发出的总功率;

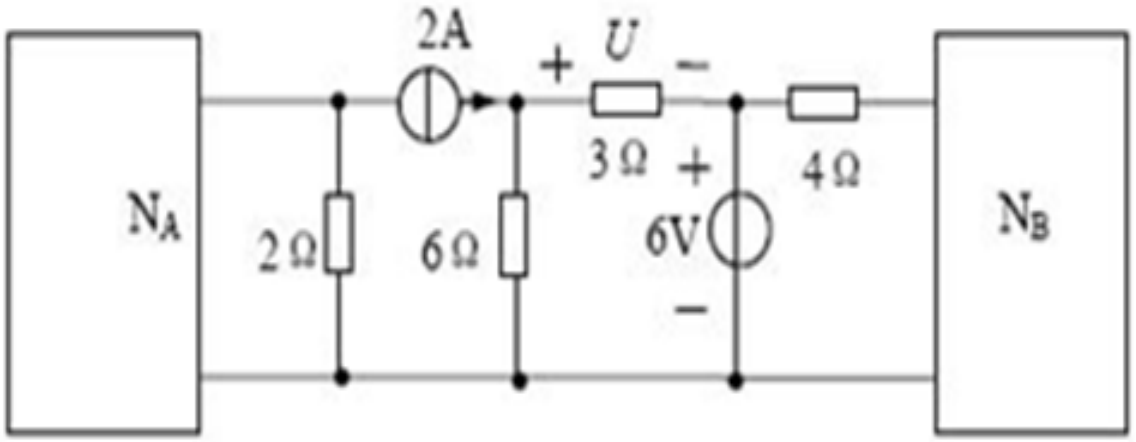
(3) 画出两表法测电动机功率的接线图, 计算功率表读数;

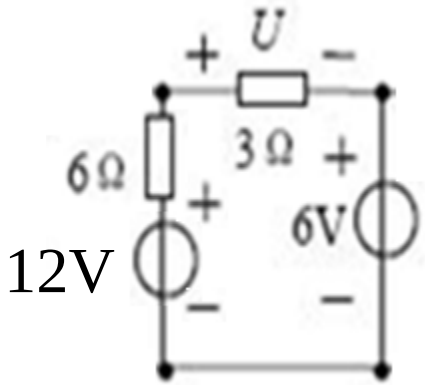
(4) 若Z为可选择的电抗，则当Z = ？时，三相总功率因数为1;

(5) 在A相接一100Ω 电阻, 重答(1). 若此时断开中线求 R 中电流.



N_A 和 N_B 均为含源线性电阻网络，在图示电路中 3 Ω 电阻的端电压 U 应为：（ ）

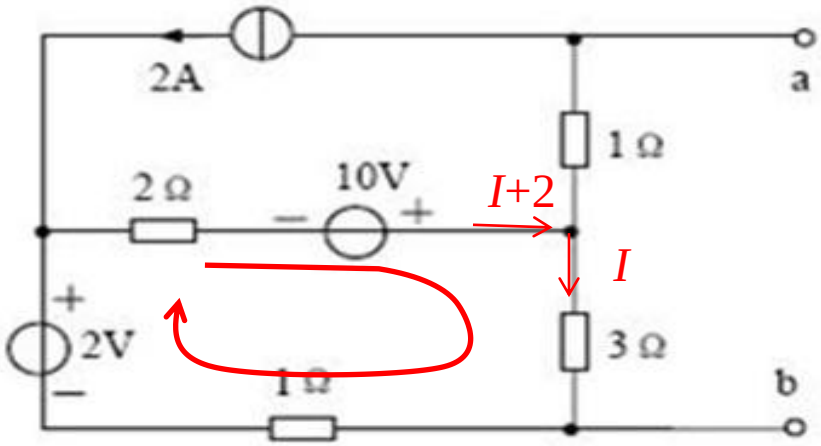




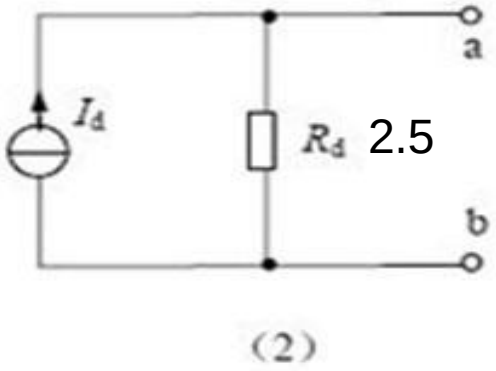
$$I_{3\Omega}=\frac{12-6}{6+3}=\frac{2}{3}$$

U=2V

电路如图（1）所示，其ab端口诺顿等效电路如图（2）所示，电路参数为：（ ）



（1）



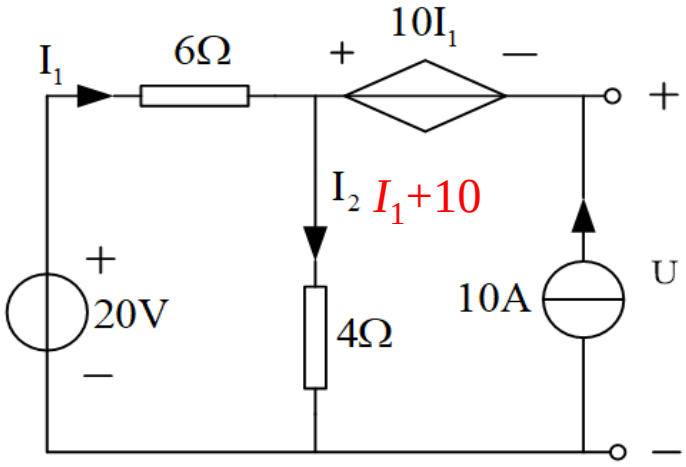
（2）

U_d=2V

R_d=3//3+1=2.5Ω

I_d=4/5A

下图中U=？

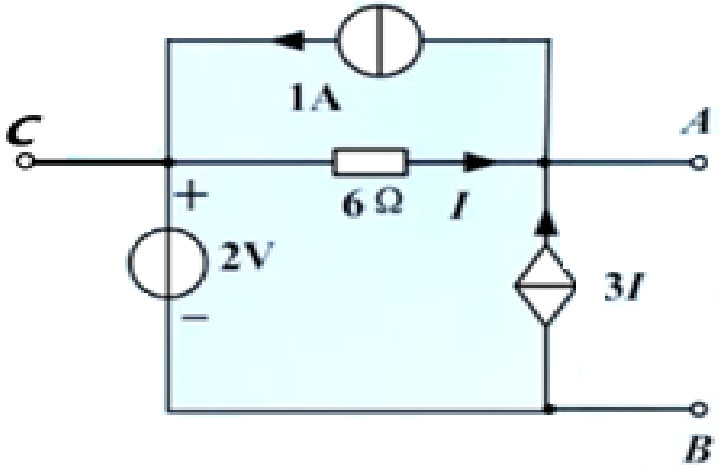


$$20=6I_1+4\left(I_1+10\right)\Rightarrow I_1=-2$$

$$U=-10I_1+4\left(I_1+10\right)$$

$$U=52V$$

如果从电压源正极处引出一条导线，端点称为C，求A、C间的戴维南等效电路。



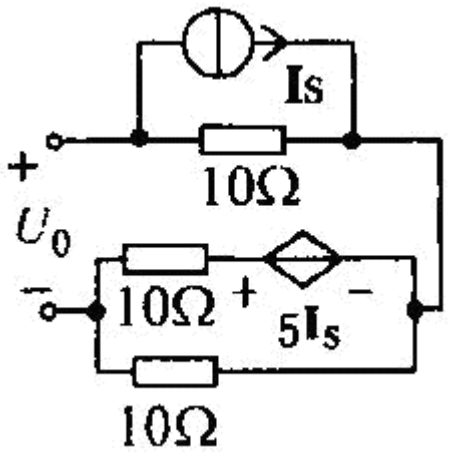
等效电压源电压

$$I+3I=1\Rightarrow I=0.25$$

$$U_d=-6I=-1.5V$$

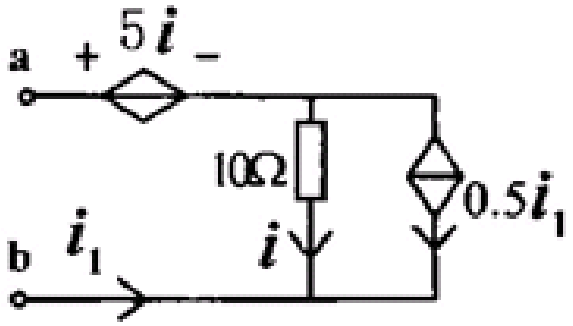
内阻 $R_d=\dfrac{U}{I}=\dfrac{6I}{I+3I}=1.5\Omega$

5. 已知I_s＝2A，求U_0



$$-10I_s-10\frac{5I_s}{10+10}=-12.5I_s=-25V$$

6. 求入端电阻

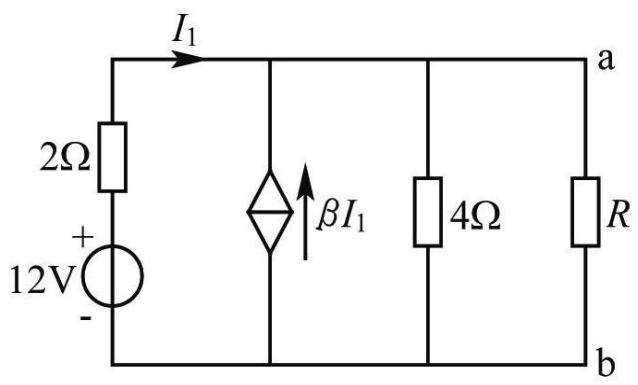


$$u_{ab}=5i+10i=15\times(-1.5i_1)$$

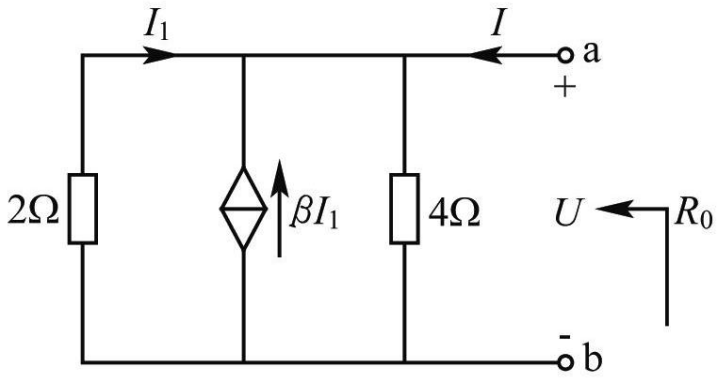
$$i_1=-0.5i_1-i$$

$$R_{ab}=22.5\Omega$$

【例】已知当R=1Ω时可获得最大电功率，求β。



(a)



(b)

图例

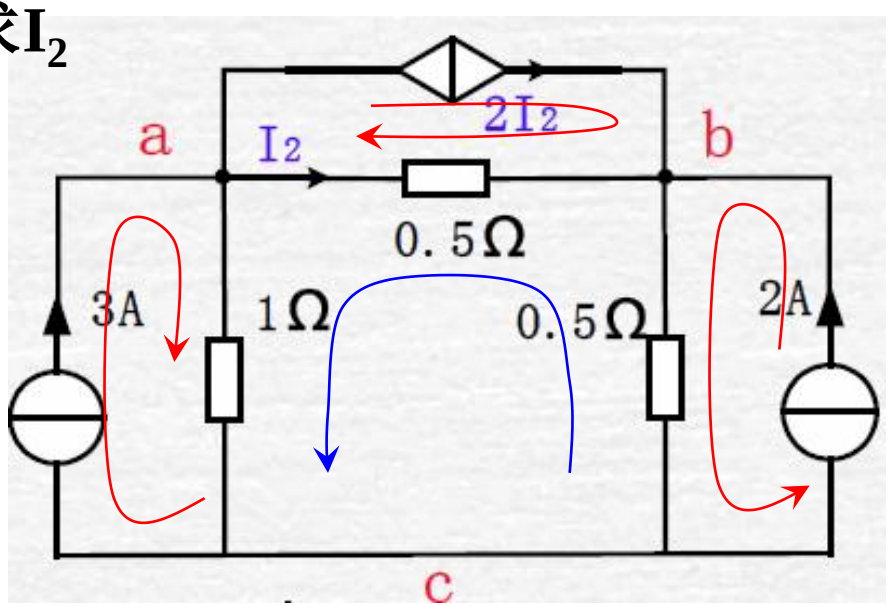
戴维南等效电阻R_0可从图 (b) 得到

$$\begin{cases} U = -2I_1 \\ U = 4(I_1 + \beta I_1 + I) \end{cases}$$

$$(3 + 2\beta)U = 4I$$

$$R_0 = \frac{U}{I} = \frac{4}{3 + 2\beta} = 1\Omega \longrightarrow \beta = 0.5$$

例2 求I_2



解法一：电源等效

$$\frac{2-1+I_2}{1+0.5+0.5}=3I_2$$

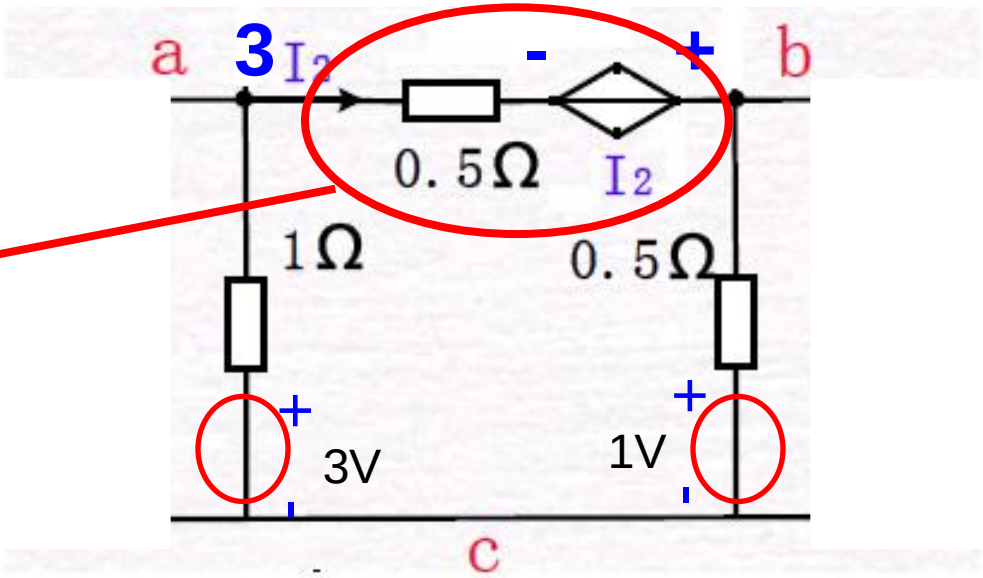
$$I_2=0.4A$$

解法二： $1\times3+0.5\times2I_2-0.5\times2-(1+0.5+0.5)\times3I_2=0$

解法三：

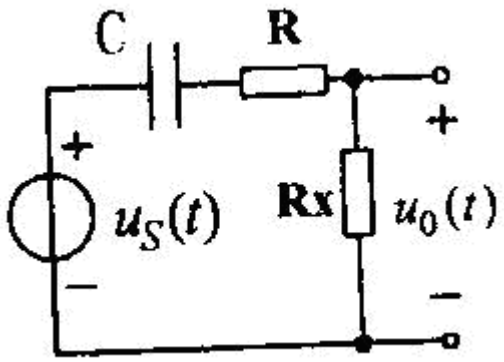
$$\frac{3-1}{1+0.5+\frac{1}{6}}=3I_2$$

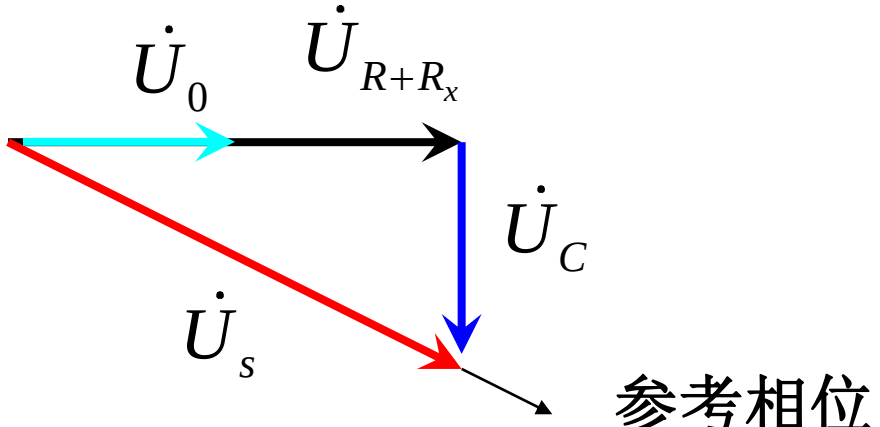
$$I_2=0.4A$$



已知Rx＝10Ω，us(t)＝10sin1000t，输出电压

u_0＝4sin(1000t＋36.9^0)，求R，C



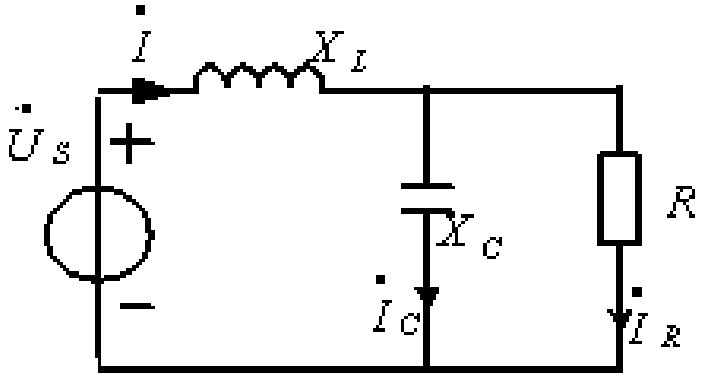


$$\dot{U}_o = \frac{\dot{U}_s}{R + R_x + \frac{1}{j\omega C}} R_x$$

$$tg\,36.9^\circ = \frac{\frac{1}{\omega C}}{R + R_x}$$

$$\left|\frac{\dot{U}_o}{\dot{U}_s}\right| = \frac{R_x}{\left|R + R_x + \frac{1}{j\omega C}\right|}$$

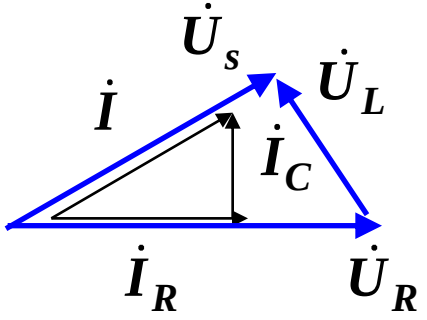
例： 已知 Us = 10v，I = 1A，I_R = 0.6A，Us 与 I 同相，试求I_C，X_L，X_C，R。



$$\cos\alpha = \frac{I_R}{I} = 0.6 \qquad \alpha = 53.1^\circ$$

$$U_{ab} = \frac{U_S}{\cos\alpha} = \frac{10}{0.6} = \frac{50}{3}(V) \qquad U_L = U_S \tan\alpha = \frac{40}{3}(V)$$

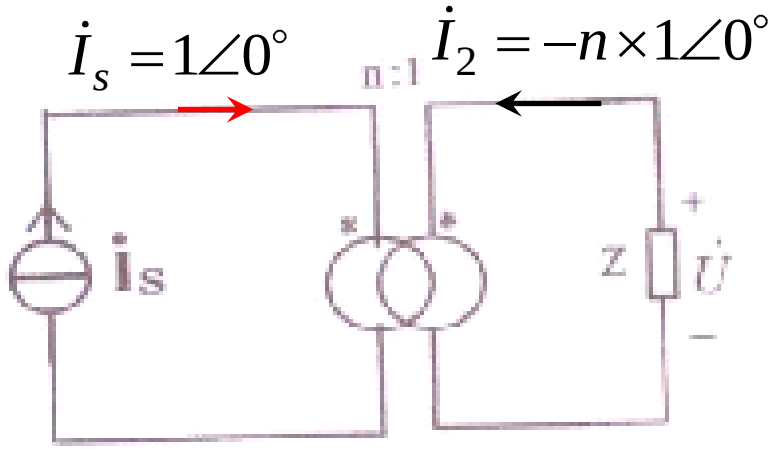
$$\dot{I}_R = 0.6\angle 0^\circ (A) \qquad \dot{I}_C = 0.8\angle 90^\circ (A)$$



$$X_L = \frac{U_L}{I} = \frac{40}{3}(\Omega) \qquad X_C = \frac{U_{ab}}{I_C} = \frac{125}{3}(\Omega)$$

$$R = \frac{U_{ab}}{I_C} = \frac{250}{9}(\Omega)$$

4. 电流源 $\dot{I}_s = 1\angle 0^\circ$ 发出的有功功率为**100W**，**n＝10**，负载端电压有效值为 $U = 10\sqrt{2}$ 求负载**Z**



Z的实部为1Ω

$$\mathrm{Re}\left\{\dot{U}_1 \dot{I}_s^*\right\} = \mathrm{Re}\left\{n\dot{U}\dot{I}_s^*\right\} = \mathrm{Re}\left\{-n\dot{I}_2 Z\dot{I}_s^*\right\} = \mathrm{Re}\left\{nnZ\dot{I}_s\dot{I}_s^*\right\} = 100$$

$$U = -I_2 Z \qquad \frac{U}{-I_2} = \sqrt{2} = Z = \sqrt{1^2 + X^2} \longrightarrow$$ **Z的虚部为1Ω**